# Disentangling Safety and Harm: Do Behavioral Signals Share Geometry Across Models?

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## Abstract

As large reasoning models become more capable, their intermediate reasoning traces expose safety-relevant decisions such as refusal, compliance, and harmful intent. This raises a central question for model safety: do their safety representations preserve a common geometric structure? We study this question through the hypothesis that safety-relevant representations contain cross-model invariants despite post-training differences. To test this, we extend two complementary methods to the cross-model setting: Activation SVD (ActSVD), which compares weight-level safety and harm subspaces, and Difference-of-Means (DoM) steering vectors, which compare activation-level behaviour directions. We further propose functional transfer and JSD convergence analyses to test whether these directions remain usable and emerge with similar layer-wise dynamics across models. We evaluate two shared-base model pairs: R1-32B vs. QwQ-32B on HarmThoughts, and Qwen3-8B vs. R1-0528-Qwen3-8B on AISafety-Student. Across both pairs, DoM directions show substantial cross-model alignment and transfer, while ActSVD subspaces drift strongly: cross-model safety, harm, and cross-type overlaps become nearly indistinguishable. These results reveal a split between what transfers and what does not: relative activation-level safety geometry is preserved, while absolute ActSVD safety/harm subspaces become model-specific after post-training. Our code is publicly available at github.com/EnzeZ1/LLM-Project.

## 1 Introduction

Large reasoning models (LRMs) produce chain-of-thought (CoT) traces that expose intermediate decisions about safety compliance (Wei et al., 2022; DeepSeek-AI et al., 2025; Korbak et al., 2025). Recent work has shown that safety-relevant features can be identified and manipulated in model activation spaces (Arditi et al., 2024; Zou et al., 2023), raising a fundamental question: *when two models share a base but undergo different post-training, how much of their safety geometry remains shared?*

We address this question through a multi-level geometric analysis (Figure 1), combining weight-level subspace decomposition (ActSVD; Wei et al., 2024) with activation-level representation analysis (DoM steering vectors, CKA, RSA) and functional transfer. Our analysis operates at four complementary levels:

1. **Weight subspaces.** Do models allocate similar singular vector subspaces to safety vs. harm activations? (ActSVD $\varphi$.)

2. **Activation geometry.** Do behaviour-level representations organise with similar pairwise structure across models? (CKA, RSA.)

3. **Functional transfer.** Can one model's DoM steering vectors discriminate behaviours in another model's activations? (Transfer AUROC.)

4. **Processing dynamics.** Do behaviours emerge at the same layers across models? (Cross-model correlation of JSD-vs-layer curves.)

We evaluate on two datasets and two corresponding model pairs. HarmThoughts (Kakkar et al., 2026) provides sentence-level harm-propagation labels (16 categories) on jailbroken traces from R1-32B and QwQ-32B, both fine-tuned from Qwen2.5-32B but via R1 distillation and reinforcement learning respectively. The AISafety-Student reasoning-safety-behaviours dataset (Menke et al., 2025) provides sentence-level annotations across 20 behavioural categories on aligned reasoning traces

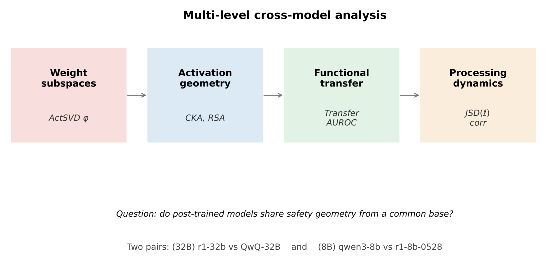


Figure 1: Four complementary levels of cross-model analysis used in this paper. Each level captures a different aspect of representational universality.

from QWEN3-8B and R1-0528-QWEN3-8B, both built on Qwen3-8B but trained via RL-based reasoning fine-tuning and R1-style distillation respectively. The two pairs thus contrast two scales (8B vs. 32B) and two post-training pipelines (distillation vs. RL) on a shared base architecture.

**Findings.** (1) DoM steering vectors share high CKA between paired models in both experiments (Figure 2); (2) ActSVD subspaces nevertheless diverge sharply, and—crucially—cross-model safety, harm, and cross-type subspace overlaps become much closer to one another than the corresponding within-model safety/harm overlaps. This suggests that post-training does not preserve a stable type-specific ActSVD matching across models (Figure 3); (3) DoM transfers across models with mean AUROC gap $\approx -0.06$ to $-0.09$ (Figure 4); (4) JSD layer-curve correlations are high (mean $r = 0.635$ and 0.693, Figure 6); (5) per-behaviour breakdown reveals that "intent-to-harmfully-comply" is the single most-reshaped representation across post-training (Figure 5).

## 2 Related Work

**Safety subspace analysis.** Wei et al. (2024) introduce ActSVD to decompose weight matrices into safety-critical and utility-critical singular-vector subspaces (we adapt this contrastive subspace view to our safety-vs.-harm setting), showing that pruning safety-only ranks ($\Delta W$) degrades safety while preserving utility. Arditi et al. (2024) identify a single "refusal direction" in activation space that causally mediates refusal behaviour.

**Representation geometry.** Centered Kernel Alignment (CKA; Kornblith et al., 2019) and Representational Similarity Analysis (RSA; Kriegeskorte et al., 2008) enable comparison of neural representations across architectures with different hidden dimensions. The Platonic Representation Hypothesis (Huh et al., 2024) posits that diverse models converge toward shared statistical representations of reality.

**Reasoning safety datasets.** Menke et al. (2025) provide step-level behavioural annotations for 20 reasoning behaviours organised into six categories (prompt interpretation, safety & risk assessment, internal cognitive process, safety-oriented response, harmful compliance, miscellaneous). Kakkar et al. (2026) introduce HARMTHOUGHTS, a benchmark of 16 sentence-level behaviour labels in jailbroken reasoning traces, spanning harmful, safety-oriented, and neutral behaviours. Lindsey et al. (2025) and Korbak et al. (2025) provide broader context for activation-level safety monitoring.

## 3 Methods

### 3.1 ActSVD: Weight-Level Subspace Analysis

Following Wei et al. (2024), for each linear projection $W$ in the transformer we collect output activations on safety-labelled data ($X_{\text{safe}}$) and harm-labelled data ($X_{\text{harm}}$). For a model $m \in \{A, B\}$ and projection $p$, we write the projected activation matrices as

$$Z^{m,p}_{\text{safe}} = W^{m,p} X^{m,p}_{\text{safe}}, \qquad Z^{m,p}_{\text{harm}} = W^{m,p} X^{m,p}_{\text{harm}}. \tag{1}$$

We then compute the rank-$r$ truncated singular value decompositions

$$Z^{m,p}_{\text{safe}} = U^{m,p}_s \Sigma^{m,p}_s (V^{m,p}_s)^\top, \qquad U^{m,p}_s \in \mathbb{R}^{d \times r}, \tag{2}$$

$$Z^{m,p}_{\text{harm}} = U^{m,p}_u \Sigma^{m,p}_u (V^{m,p}_u)^\top, \qquad U^{m,p}_u \in \mathbb{R}^{d \times r}. \tag{3}$$

The columns of $U^{m,p}_s$ and $U^{m,p}_u$ define the safety and harm activation subspaces for model $m$ and projection $p$. Throughout we use $r = 100$ and sample $N_{\text{safe}} = N_{\text{harm}} = 128$ examples per partition.

**Subspace similarity.** For two orthonormal subspace bases $U \in \mathbb{R}^{d \times r_U}$ and $V \in \mathbb{R}^{d \times r_V}$, we measure overlap by the normalized projection affinity

$$\varphi(U, V) = \frac{\|U^\top V\|_F^2}{\min(r_U, r_V)} = \frac{\operatorname{tr}(UU^\top VV^\top)}{\min(r_U, r_V)}. \tag{4}$$

Equivalently, if $\{\theta_k\}_{k=1}^{\min(r_U,r_V)}$ are the principal angles between the two subspaces, then

$$\varphi(U,V) = \frac{1}{\min(r_U,r_V)} \sum_{k=1}^{\min(r_U,r_V)} \cos^2\theta_k. \quad (5)$$

Thus $\varphi \in [0,1]$, where lower values indicate greater subspace separability.

For each projection we compute the within-model overlaps

$$\varphi_{su}^{A,p} = \varphi(U_s^{A,p}, U_u^{A,p}), \qquad \varphi_{su}^{B,p} = \varphi(U_s^{B,p}, U_u^{B,p}), \quad (6)$$

and the cross-model overlaps

$$\varphi_{ss}^{p} = \varphi(U_s^{A,p}, U_s^{B,p}), \quad (7)$$

$$\varphi_{uu}^{p} = \varphi(U_u^{A,p}, U_u^{B,p}), \quad (8)$$

$$\varphi_{s\to u}^{p} = \varphi(U_s^{A,p}, U_u^{B,p}). \quad (9)$$

The contrast between the within-model quantities $\varphi_{su}^{A,p}, \varphi_{su}^{B,p}$ and the cross-model quantities $\varphi_{ss}^{p}, \varphi_{uu}^{p}, \varphi_{s\to u}^{p}$ measures how much the safety/harm distinction is preserved across models.

**Label partitioning.** For HARMTHOUGHTS, safety labels are {RA, AL, CC, IA, FL, ED} (refusal/alignment/etc.) and harm labels are {RS, CR, PS, IR, PA, TD, DKE, HV, CE, OB} (harm-execution behaviours), following Kakkar et al. (2026). For AISAFETY-STUDENT we contrast the harmful-compliance category (Group V: `V_INTEND_HARMFUL_COMPLIANCE`, `V_DETAIL_HARMFUL_METHOD_OR_INFO`, `V_NOTE_RISK_WHILE_DETAILING_HARM`) against all remaining 17 behaviours as a non-harmful-compliance contrast partition.

### 3.2 DoM: Activation-Level Steering Vectors

For each behaviour $b$, layer $\ell$, and component $c \in \{\mathrm{attn}, \mathrm{mlp}, \mathrm{residual}\}$, let $P_b$ and $N_b$ denote the behaviour-positive and behaviour-negative contrastive training sets. We first compute the positive and negative activation centroids

$$\mu_{b,+}^{\ell,c} = \frac{1}{|P_b|} \sum_{i\in P_b} h_i^{\ell,c}, \quad (10)$$

$$\mu_{b,-}^{\ell,c} = \frac{1}{|N_b|} \sum_{j\in N_b} h_j^{\ell,c}. \quad (11)$$

The Difference-of-Means steering vector is then

$$v_b^{\ell,c} = \mu_{b,+}^{\ell,c} - \mu_{b,-}^{\ell,c} = \frac{1}{|P_b|} \sum_{i\in P_b} h_i^{\ell,c} - \frac{1}{|N_b|} \sum_{j\in N_b} h_j^{\ell,c}. \quad (12)$$

For transfer and scoring experiments, we use the unit-normalized direction

$$\hat{v}_b^{\ell,c} = \frac{v_b^{\ell,c}}{\|v_b^{\ell,c}\|_2 + \varepsilon}, \quad (13)$$

where $\varepsilon$ is a small numerical stabilizer. Positive and negative examples are drawn from disjoint prompt-level train/test splits (80/20). For activation extraction we hook the output of `self_attn.o_proj` ("attn"), the MLP block output ("mlp"), and the entire decoder block output ("residual").

### 3.3 Cross-Model Geometry

**CKA.** Given behaviour-level DoM steering directions $X \in \mathbb{R}^{n\times d_A}$ and $Y \in \mathbb{R}^{n\times d_B}$, where each row corresponds to the unit-normalized DoM direction for one shared behaviour, we first center the representations by

$$H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad X_c = HX, \qquad Y_c = HY. \quad (14)$$

The centered Gram matrices are

$$K_X = X_c X_c^\top, \qquad K_Y = Y_c Y_c^\top. \quad (15)$$

We compute linear CKA as

$$\mathrm{CKA}(X,Y) = \frac{\langle K_X, K_Y\rangle_F}{\sqrt{\langle K_X, K_X\rangle_F \langle K_Y, K_Y\rangle_F}}. \quad (16)$$

Equivalently, in feature space,

$$\mathrm{CKA}(X,Y) = \frac{\|X_c^\top Y_c\|_F^2}{\|X_c^\top X_c\|_F \|Y_c^\top Y_c\|_F}. \quad (17)$$

CKA is invariant to orthogonal rotations and isotropic scaling, and is well-defined when $d_A \neq d_B$.

**RSA.** For each model, we build a cosine-distance Representational Dissimilarity Matrix (RDM) over behaviour-level DoM directions. For normalized directions $\{\hat{v}_i\}_{i=1}^n$, the RDM entries are

$$D_{ij} = 1 - \frac{\langle \hat{v}_i, \hat{v}_j\rangle}{\|\hat{v}_i\|_2 \|\hat{v}_j\|_2}. \quad (18)$$

Let

$$d^A = \mathrm{vec}_{i<j}(D_{ij}^A), \qquad d^B = \mathrm{vec}_{i<j}(D_{ij}^B). \quad (19)$$

We report RSA as the Spearman rank correlation

$$\mathrm{RSA}(A,B) = \rho_{\mathrm{Spearman}}(d^A, d^B). \quad (20)$$

### 3.4 Functional Transfer (Ours)

We test whether model $A$'s DoM directions can detect behaviours in model $B$'s activations. For a test activation $h_{i,B}^{\ell,c}$ from model $B$, we compute the cross-model projection score

$$s_{i,b}^{A\to B,\ell,c} = \langle h_{i,B}^{\ell,c}, \hat{v}_{b,A}^{\ell,c} \rangle. \tag{21}$$

The within-model baseline uses the target model's own direction:

$$s_{i,b}^{B\to B,\ell,c} = \langle h_{i,B}^{\ell,c}, \hat{v}_{b,B}^{\ell,c} \rangle. \tag{22}$$

For each behaviour $b$, AUROC is computed as

$$\mathrm{AUROC}_{A\to B}^{b,\ell,c} = \Pr\left[ s_{+,b}^{A\to B,\ell,c} > s_{-,b}^{A\to B,\ell,c} \right]. \tag{23}$$

Averaging over behaviours gives the transfer gap

$$\Delta_{\mathrm{AUROC}}^{A\to B} = \frac{1}{|\mathcal{B}|} \sum_{b\in\mathcal{B}} \Delta_b^{A\to B}, \tag{24}$$

where

$$\Delta_b^{A\to B} = \mathrm{AUROC}_{A\to B}^{b} - \mathrm{AUROC}_{B\to B}^{b}. \tag{25}$$

The two models must share hidden dimension for transfer to be directly defined; both of our pairs satisfy this condition (32B: $d = 5120$; 8B: $d = 4096$).

### 3.5 JSD Convergence (Ours)

For each behaviour $b$ and layer $\ell$, we project positive and negative activations onto the corresponding steering direction:

$$z_{i,b}^{\ell} = \langle h_i^{\ell}, \hat{v}_b^{\ell} \rangle. \tag{26}$$

We discretize the positive and negative projected scores into 50-bin empirical distributions $P_b^{\ell}$ and $Q_b^{\ell}$, and define their mixture distribution as

$$M_b^{\ell} = \frac{1}{2}\left( P_b^{\ell} + Q_b^{\ell} \right). \tag{27}$$

The Jensen–Shannon divergence is

$$\mathrm{JSD}_b^{\ell} = \frac{1}{2} D_{\mathrm{KL}}\big(P_b^{\ell} \,\|\, M_b^{\ell}\big) + \frac{1}{2} D_{\mathrm{KL}}\big(Q_b^{\ell} \,\|\, M_b^{\ell}\big). \tag{28}$$

The KL term is computed as

$$D_{\mathrm{KL}}(P\|M) = \sum_{k=1}^{50} P_k \log \frac{P_k + \varepsilon}{M_k + \varepsilon}, \tag{29}$$

where $\varepsilon$ is used only for numerical stability.

## 4 Experimental Setup

### 4.1 Models

**32B pair: R1-32B vs. QwQ-32B.** Both fine-tuned from the Qwen2.5-32B base. R1-32B (*deepseek-ai/DeepSeek-R1-Distill-Qwen-32B*) is produced by distilling reasoning traces from DeepSeek-R1; QwQ-32B (*Qwen/QwQ-32B*) is produced via reinforcement-learning-based reasoning fine-tuning. Both have $d_{\text{hidden}} = 5120$ and 64 layers.

**8B pair: QWEN3-8B vs. R1-0528-QWEN3-8B.** Both built from the Qwen3-8B base. QWEN3-8B (*Qwen/Qwen3-8B*) is the official RL-aligned reasoning release, while R1-0528-QWEN3-8B (*deepseek-ai/DeepSeek-R1-0528-Qwen3-8B*) is an R1-style distilled variant. Both have $d_{\text{hidden}} = 4096$ and 36 layers.

### 4.2 Datasets

**HARMTHOUGHTS.** We use the *ishitakakkar-10/HarmThoughts* dataset, restricted to entries labelled with R1-32B or QwQ-32B as source model. Sentences are tagged with one of 16 harm-propagation behaviours. After prompt-level 80/20 splitting we obtain ∼2,000 contrastive train pairs per behaviour for DoM and $N = 128$ samples per partition for ActSVD.

**AISAFETY-STUDENT.** The *AISafety-Student/reasoning-safety-behaviours* dataset (Menke et al., 2025) contains 22,809 sentence-level annotations across 20 behaviour labels, of which 13,523 are tagged QWEN3-8B and 9,286 are tagged R1-0528-QWEN3-8B. After prompt-level 80/20 splitting and one-vs-rest contrastive sampling, the train splits contain 11,040 and 13,602 constructed contrastive pairs, respectively.

### 4.3 Compute

ActSVD and DoM are extracted on NVIDIA A40 GPUs (8×48 GB) with `device_map="auto"` pipeline parallelism in `transformers`. Extraction throughput is roughly 18 samples/s for the 8B models and 2.5 samples/s for the 32B models on this hardware.

## 5 Results

### 5.1 DoM Steering Vectors Share Cross-Model Structure

Figure 2 and Table 1 show that DoM steering vectors are strongly aligned between paired models.

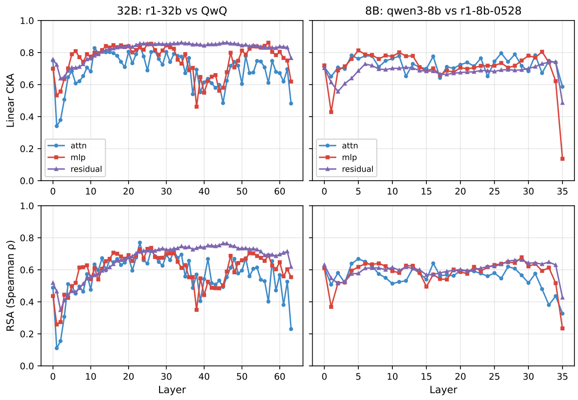


Figure 2: Layer-wise CKA (top) and RSA (bottom) of DoM steering vectors between paired models, decomposed by component (attn / mlp / residual). Both pairs maintain CKA above $\sim 0.6$ across most layers, with small dips around boundary layers; the 8B pair plateaus at $\sim 0.7$, while the 32B pair shows a residual-stream peak around layers 20–35 (CKA $\approx 0.85$).

| **Pair** | **CKA attn** | **RSA attn** | **CKA mlp** | **RSA mlp** |
|---|---|---|---|---|
| 32B (r1 vs. QwQ) | 0.691 | 0.564 | 0.757 | 0.601 |
| 8B (qwen3 vs. R1-0528) | 0.724 | 0.559 | 0.709 | 0.582 |
| | | **residual** | | |
| 32B | CKA = 0.819 | | RSA = 0.666 | |
| 8B | CKA = 0.680 | | RSA = 0.601 | |

Table 1: Mean linear CKA and RSA of DoM steering vectors averaged over layers.

CKA remains above 0.6 across most mid-layers, with the 32B pair reaching 0.819 on the residual stream. RSA shows the same qualitative trend at slightly lower magnitudes. This indicates that the relative geometry of behaviour-level directions is partially preserved across post-training.

### 5.2 ActSVD Subspaces Diverge Across Models

Table 2 and Figure 3 show the main contrast with DoM. Within each model, safety and harm subspaces remain partially separable. Across models, however, safety, harm, and cross-type overlaps become nearly indistinguishable ($\varphi_{ss} \approx \varphi_{uu} \approx \varphi_{s \to u}$). Thus, the safety/harm distinction does not survive as an absolute weight-level subspace across post-training.

### 5.3 DoM Vectors Transfer Functionally

Table 3 and Figure 4 show that DoM directions remain functionally usable across models. Mean transfer gaps are small: $-0.058$ for the 32B pair

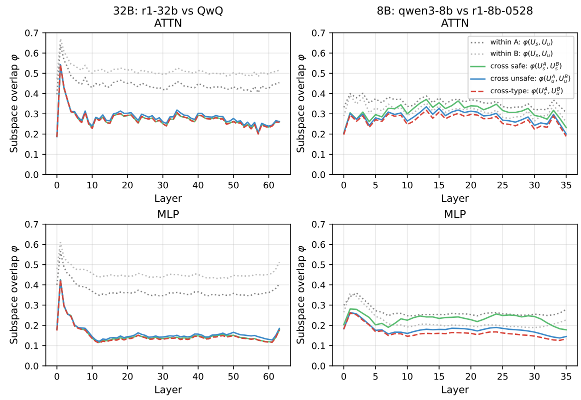


Figure 3: Layer-wise ActSVD subspace overlaps. Within-model $\varphi(U_s, U_u)$ is higher than all cross-model quantities, while cross-safe, cross-unsafe, and cross-type overlaps collapse to nearly the same value.

| **Quantity** | **32B** | **8B** |
|---|---|---|
| *Within-model*, attention | | |
| $\varphi^A_{su}$ | 0.445 | 0.355 |
| $\varphi^B_{su}$ | 0.512 | 0.315 |
| *Within-model*, MLP | | |
| $\varphi^A_{su}$ | 0.369 | 0.266 |
| $\varphi^B_{su}$ | 0.455 | 0.218 |
| *Cross-model*, attention | | |
| $\varphi(U^A_s, U^B_s)$ | 0.276 | 0.313 |
| $\varphi(U^A_u, U^B_u)$ | 0.284 | 0.282 |
| $\varphi(U^A_s, U^B_u)$ | 0.275 | 0.268 |
| *Cross-model*, MLP | | |
| $\varphi(U^A_s, U^B_s)$ | 0.153 | 0.232 |
| $\varphi(U^A_u, U^B_u)$ | 0.161 | 0.180 |
| $\varphi(U^A_s, U^B_u)$ | 0.150 | 0.164 |

Table 2: ActSVD subspace overlap statistics averaged over projections.

and $-0.079$ for the 8B pair. This supports the hypothesis that activation-level safety directions contain cross-model invariants, even when ActSVD subspaces drift.

### 5.4 JSD Dynamics and Behaviour-Level Asymmetry

Cross-model JSD curve correlations average $r = 0.635$ for the 32B pair and $r = 0.693$ for the 8B pair (Figure 6). For the 32B pair, the strongest agreements occur for harm-execution behaviours such as `OB` and `IA` ($r > 0.8$), while the weakest agreement is on `CR` ($r = 0.38$). For the 8B pair, the highest correlations occur for fact/knowledge statements, immediate reasoning-step planning, and harmful-method detailing ($r = 0.896$, 0.891, and 0.890). The weakest cases are self-correction and harmful-compliance intent, with correlations

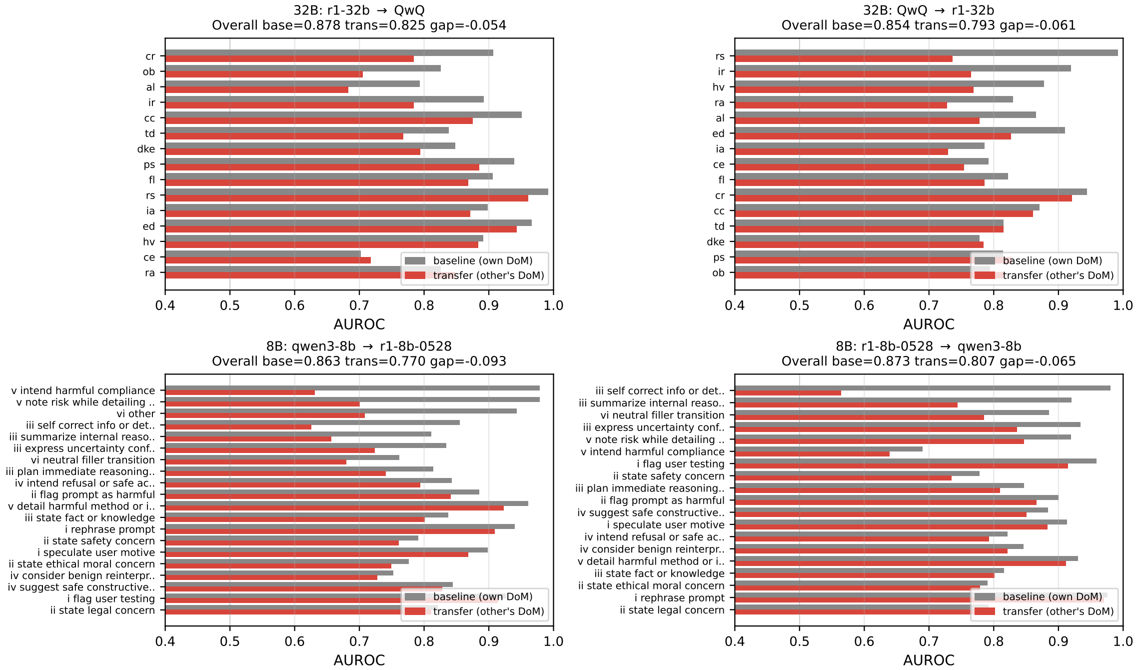


Figure 4: Per-behaviour transfer AUROC for all four directions. Grey bars are within-model baselines; red bars use transferred DoM directions.

| Direction | Base | Trans. | Gap |
|---|---|---|---|
| 32B: r1-32b $\rightarrow$ QwQ | 0.878 | 0.825 | $-0.054$ |
| 32B: QwQ $\rightarrow$ r1-32b | 0.854 | 0.793 | $-0.061$ |
| 8B: qwen3-8b $\rightarrow$ R1-0528 | 0.863 | 0.770 | $-0.093$ |
| 8B: R1-0528 $\rightarrow$ qwen3-8b | 0.873 | 0.807 | $-0.065$ |
| 32B mean | 0.866 | 0.809 | $-0.058$ |
| 8B mean | 0.868 | 0.789 | $-0.079$ |

Table 3: Transfer AUROC: source-model DoM applied to target-model activations.

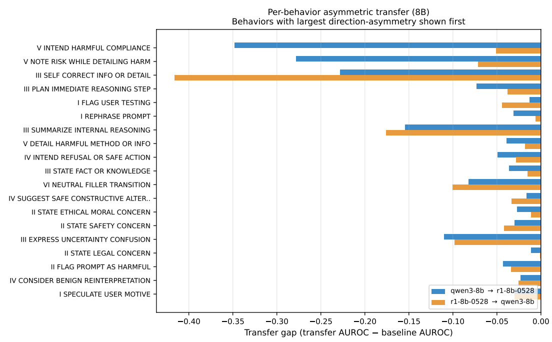


Figure 5: Per-behaviour transfer gap in the 8B pair, sorted by direction asymmetry. Harmful-compliance intent shows the largest asymmetry, with gap $-0.348$ in the qwen3 $\rightarrow$ r1 direction versus $-0.051$ in the reverse direction.

$r = -0.226$ and $r = 0.073$.

Figure 5 shows that harmful-compliance intent is the most asymmetric behaviour in the 8B pair: the qwen3 $\rightarrow$ r1 direction has gap $-0.348$, while the reverse direction has gap only $-0.051$. This matches the JSD result for the same behaviour, whose layer-wise correlation is almost zero ($r = 0.073$). Together, these results suggest that R1-style distillation substantially reshapes this decision-related representation.

By contrast, legal, ethical, and safety-concern statements transfer almost losslessly and have high JSD correlations. This suggests that recognition-style behaviours remain closer to the shared base, while decision-style behaviours are more strongly reshaped by post-training.

## 5.5 Cross-Experiment Summary

Across both model pairs, the qualitative pattern is consistent. DoM directions preserve activation-level structure across post-training, as shown by high CKA/RSA, small transfer AUROC gaps, and correlated JSD layer dynamics. ActSVD, however, reveals that absolute weight-level subspaces are less stable: safety, harm, and cross-type cross-model overlaps collapse to nearly the same value. This supports the main conclusion that post-training preserves relative activation-level geometry while reshaping absolute safety subspaces.

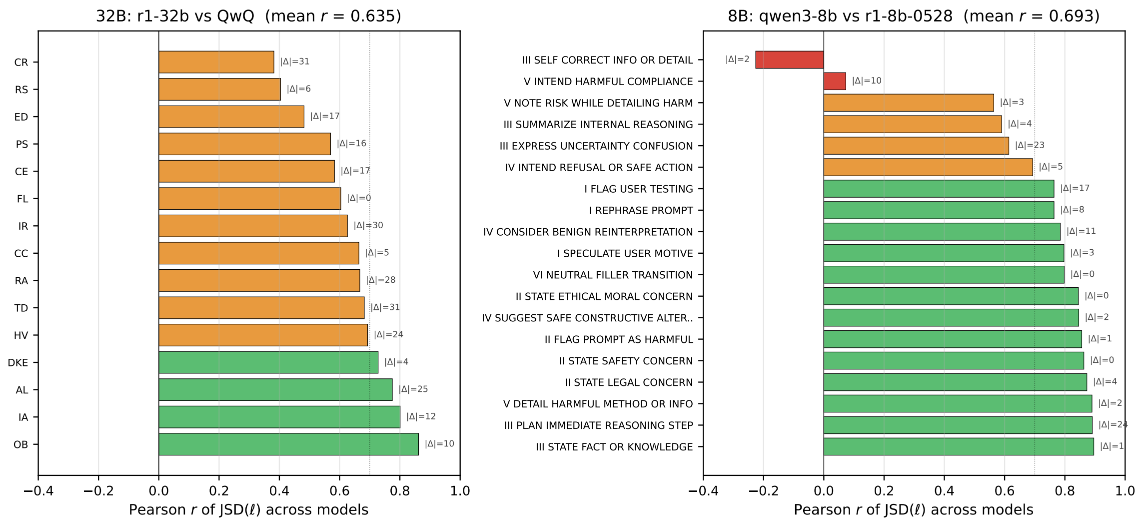


Figure 6: Per-behaviour Pearson correlation of layer-wise JSD curves between paired models.

## 6 Discussion and Conclusion

Across two shared-base model pairs, we find a consistent separation between activation-level and weight-level geometry. DoM steering vectors remain aligned and functionally transferable across post-training, while ActSVD subspaces drift: cross-model safety, harm, and cross-type overlaps collapse to nearly the same value. This suggests that post-training preserves relative activation-level behaviour geometry, but reshapes absolute weight-level safety subspaces.

One interpretation is that post-training applies a roughly global rotation or reparameterization of the activation space. Such a transformation can preserve relative geometry, which CKA and DoM transfer can still detect, while destroying direct coordinate-level alignment between ActSVD subspaces. This explains why activation-space tools transfer more reliably than weight-space interventions: DoM directions retain functional signal, whereas ActSVD safety/harm subspaces must be recomputed per model.

The behaviour-level analysis further shows that not all safety behaviours are equally stable. Recognition-style behaviours, such as legal, ethical, and safety-concern statements, transfer cleanly and have highly correlated JSD dynamics. In contrast, decision-style behaviours, especially harmful-compliance intent, show strong directional asymmetry. This supports the view that post-training most strongly reshapes representations tied to safety decisions rather than factual or normative recognition.

Overall, our results support a qualified version of the shared safety geometry hypothesis. Safety representations are not universal as absolute weight-level subspaces, but they do retain substantial activation-level structure across models sharing a base architecture. This distinction is important for safety tooling: activation-level probes may generalize across post-trained variants, while weight-level interventions likely require model-specific recalibration.

**Limitations and future work.** Our analysis is limited to two model pairs and two sentence-level safety datasets. The datasets also use different taxonomies and prompt distributions. Some rare behaviours, especially harmful-compliance intent, have few examples and should be validated on larger annotated corpora. Future work should test more model families, more post-training pipelines, and causal interventions based on the transferred DoM directions. More broadly, an important open question is whether safety-related subspaces eventually converge under larger scale, shared pretraining, or similar alignment objectives, or whether post-training consistently makes absolute safety/harm subspaces model-specific while preserving only relative activation-level geometry.

## 7 Group Contributions

**Enze Zhang.** Proposed the core idea of the project, organized the CKA, RSA, and DoM analysis, and designed the functional-transfer and JSD-convergence experiments. Implemented the transfer and JSD-convergence methods (25%).

**Youran Wang.** Implemented the CKA and RSA methods, contributed to cross-model geometry analysis, and created plotting scripts for the representation-geometry experiments (25%).

**Sungjin Cheong.** Implemented the DoM and ActSVD pipelines, collected and organized the experimental code, and ran the main experiments (25%).

**Yuliang Wu.** Implemented data processing for the two datasets, including splitting examples into safety and harm partitions, and contributed to cross-model analysis (25%).

**Everyone contributed to this report equally.**

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## A Additional Implementation Details

All experiments use prompt-level 80/20 train/test splits so that sentences from the same reasoning trace do not appear in both training and test sets. For ActSVD, we construct safety and harm activation partitions and compute the top $r = 100$ left singular vectors of the projected activation matrices. We sample $N_{\text{safe}} = N_{\text{harm}} = 128$ examples per partition.

For DoM, behaviour-positive and behaviour-negative contrastive examples are used to construct steering vectors. Each DoM vector is normalized before scoring:

$$\hat{v}_b^{\ell,c} = \frac{v_b^{\ell,c}}{\|v_b^{\ell,c}\|_2 + \varepsilon}. \tag{30}$$

Transfer is evaluated only between model pairs with the same hidden dimension, so that the source model's DoM direction can be applied directly to the target model's activations. For JSD analysis, projected positive and negative activation scores are discretized into 50-bin empirical distributions before computing layer-wise Jensen–Shannon divergence curves.

## B Additional Notes on Behaviour Labels

Some behaviour labels in AISafety-Student are long, such as harmful-compliance intent, self-correction, and harmful-method detailing. In the main text, we use shortened natural-language descriptions to avoid overfull lines in the ACL two-column format. The complete labels are shown in the corresponding figures.